A context-aware framework leveraging LLM-estimated variant effects for individualized variant-to-phenotype interpretation

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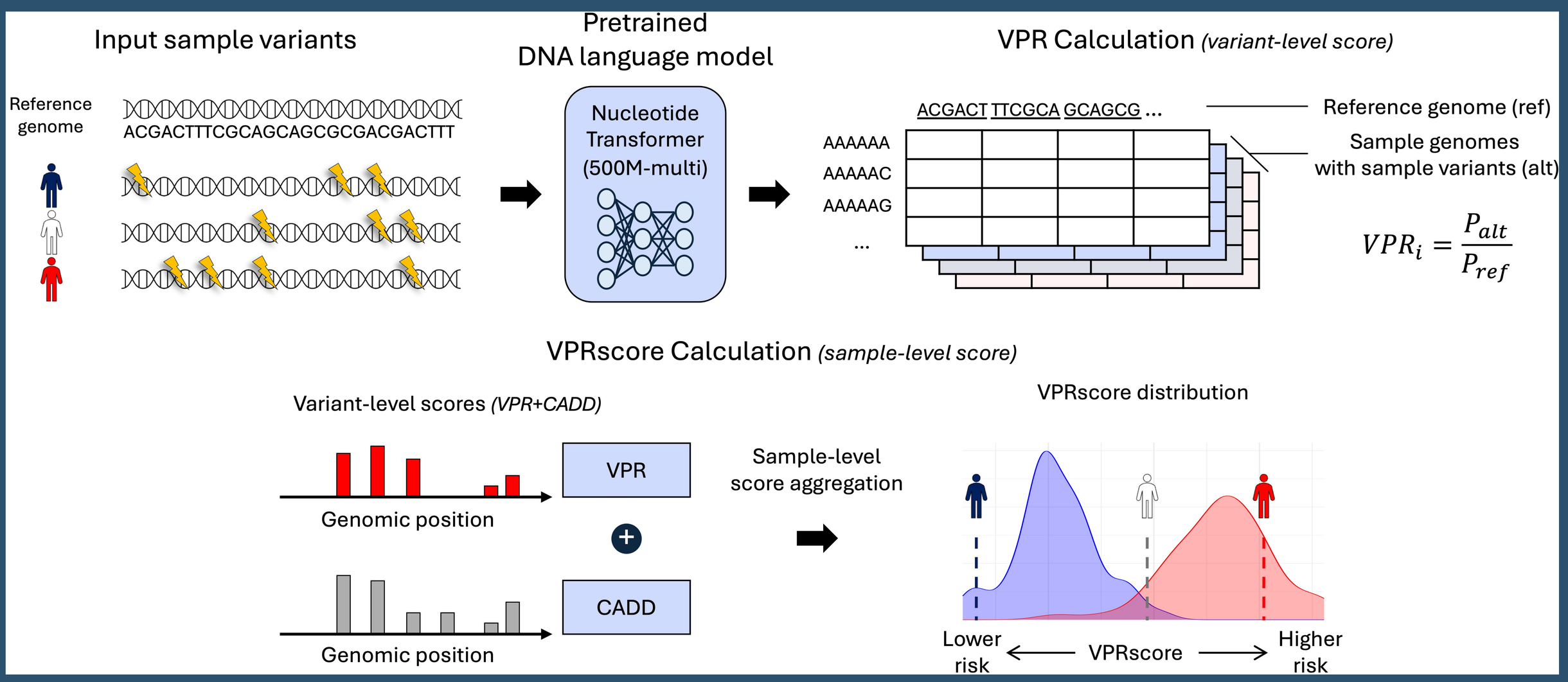
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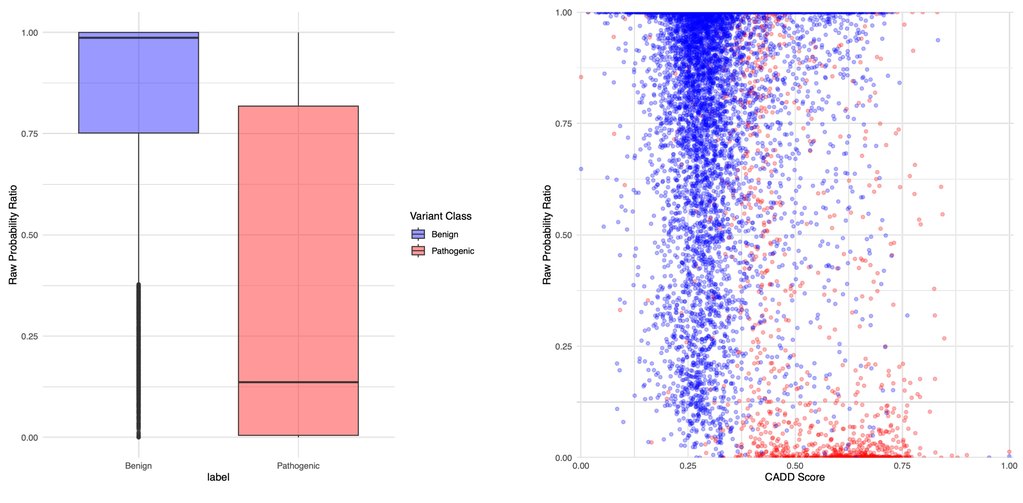
Background

- Polygenic risk scores (PRS) rely on GWAS-derived, population-level effect sizes, limiting transferability across ancestries and the representation of rare or individual-specific variants.
- Genomic language models can estimate variant effects directly from local DNA sequence context, offering an alternative source of functional information.
- Goal: develop a context-aware framework that converts sequence-derived variant effects into an individual-level genetic risk score.

VPRscore Framework

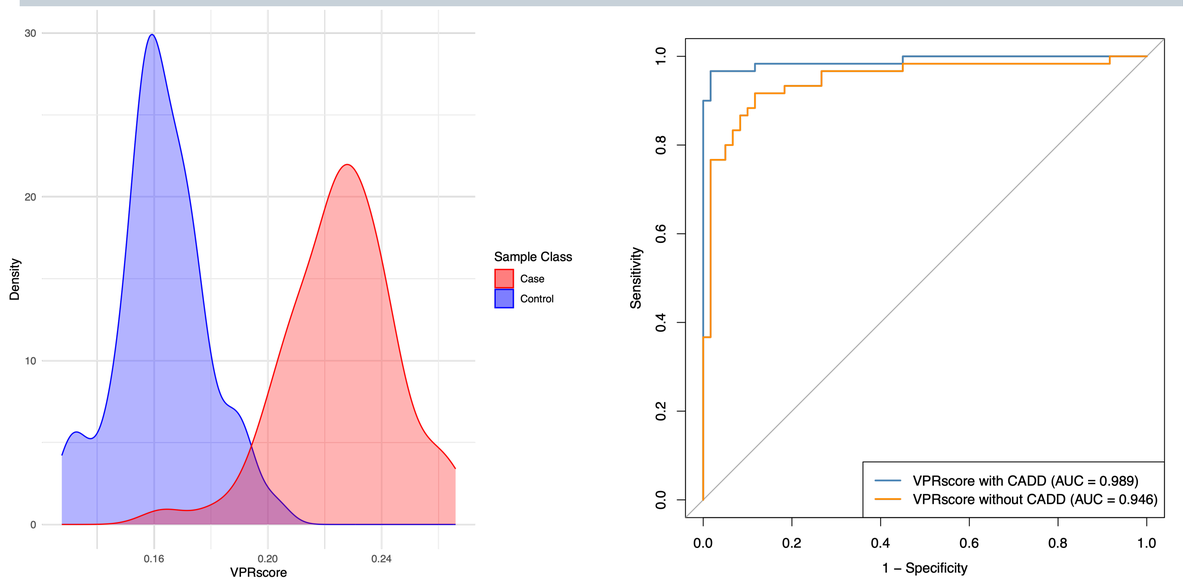


Variant-level VPR captures pathogenicity



- Lower VPRs in pathogenic variants
- Modest VPR–CADD correlation
  → Complementary signals

VPRscore for sample-level risk stratification



- Clear case–control separation (**AUROC = 0.989**)
- CADD improves performance: **0.946 → 0.989**

Take-home Message

- VPRscore translates sequence-context effects into individualized pathogenic burden.
- Independent real-world validation remains needed.

Simulated case–control cohort design

	Diseases Variants	Diseases-Associated		Non-Diseases		Total Count	Average CADD
		Patho-genic	Benign	Patho-genic	Benign		
Case	35~40	25~30	35~40	25~30	25~30	145~180	27.92
Control	15~20	5~10	35~40	25~30	60~70	140~180	18.85

Acknowledgements

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